

Introduction to Communication Theory

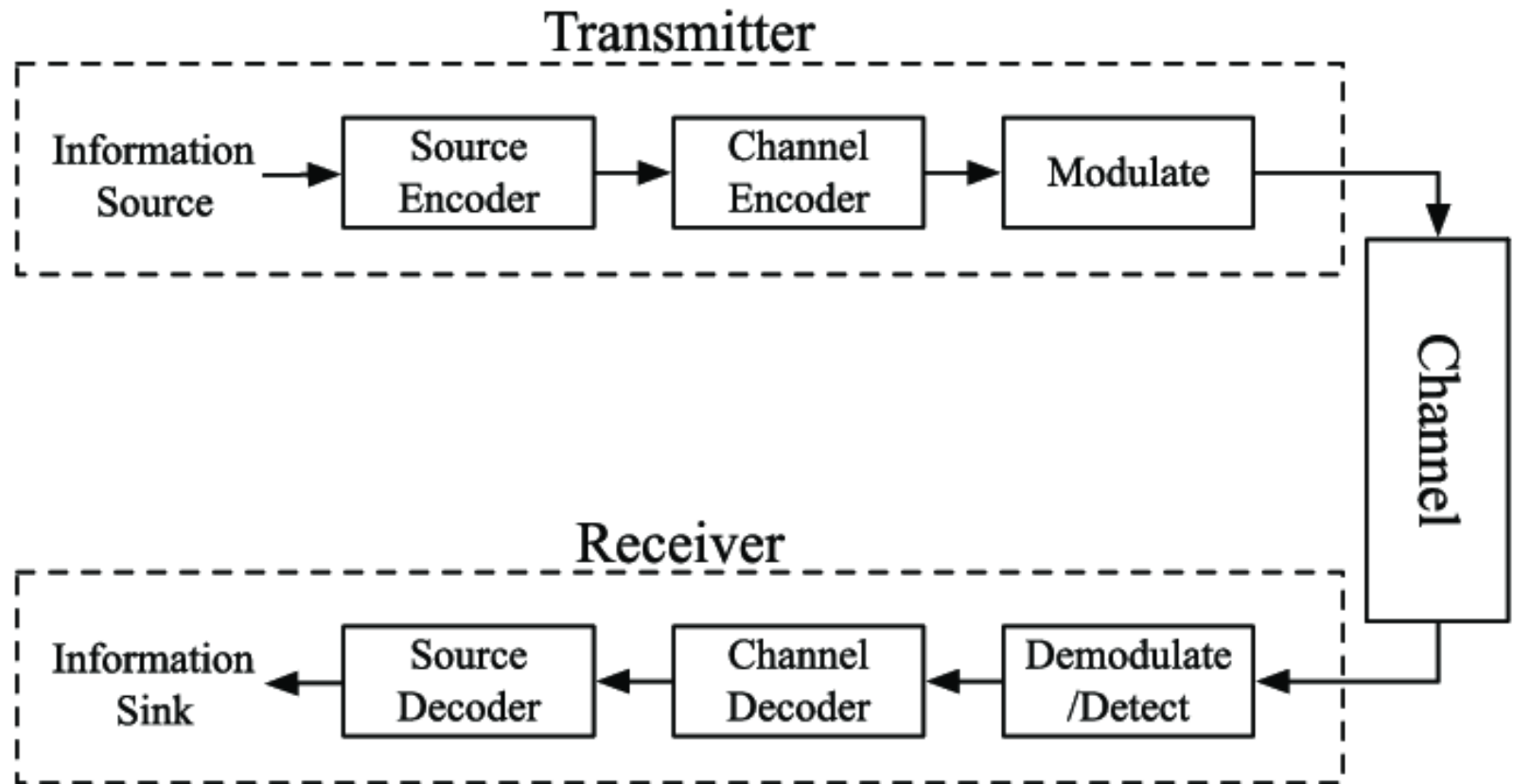
Introduction

- ❑ Long distance communication is called **Telecommunication**
- ❑ Technically Communication is the summation of:
 - **Transmission**
 - **Reception and**
 - **Processing of information by electric means**
- ❑ Radio communication was greatly improved during **2nd world war**

Examples of Communication System

- **Telephony and Telegraphy**
- **Radio Broadcasting (Example?)**
- **Point-to-point & Mobile Communication**
- **Computer Communication**
- **Radar (Applications?)**
- **Television Broadcasting**
- **Radio Telemetry**
- **Radio aids to Navigation**
- **Radio aids to Aircraft Landing etc.**

Block Diagram of General Communication System(1/3)



Block Diagram of General Communication System(1/3)

Source Encoder

The source encoder compresses the data into minimum number of bits. This process helps in effective utilization of the bandwidth. It removes the redundant bits (unnecessary excess bits, i.e., zeroes).

Channel Encoder

The channel encoder, does the coding for error correction. During the transmission of the signal, due to the noise in the channel, the signal may get altered and hence to avoid this, the channel encoder adds some redundant bits to the transmitted data. These are the error correcting bits.

Block Diagram of General Communication System(2/3)

Channel Decoder

The channel decoder, after detecting the sequence, does some error corrections. The distortions which might occur during the transmission, are corrected by adding some redundant bits. This addition of bits helps in the complete recovery of the original signal.

Source Decoder

The resultant signal is once again digitized by sampling and quantizing so that the pure digital output is obtained without the loss of information. The source decoder recreates the source output.

Data and Signal

❑ Data

- Entities that convey meaning

❑ Signals

- Electric or electromagnetic representations of data, physically propagates along medium

❑ Transmission

- Communication of data by propagation and processing of signals

❑ Coding: Data -> Signal

- How is information coded in a format suitable for transmission?

Signal Representation

❑ Time domain concepts

➤ Analog signal

- Varies in a smooth way over time

➤ Digital signal

- Maintains a constant level then changes to another constant level

➤ Periodic signal

- Pattern repeated over time

➤ Aperiodic signal

- Pattern not repeated over time

Spectrum & Bandwidth

- ❑ Spectrum

- range of frequencies contained in signal

- ❑ Absolute bandwidth

- width of spectrum

- ❑ Effective bandwidth

- Often just bandwidth
 - Narrow band of frequencies containing most of the energy

- ❑ DC Component

- Component of zero frequency

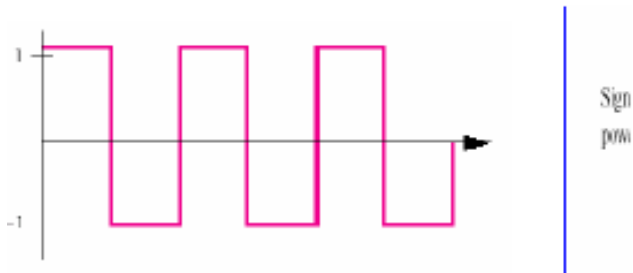
Data Rate and Bandwidth

- ❑ **Definition:** Bandwidth is the maximum potential ("width of the pipe"), while data rate is the actual speed ("flow of water").
- ❑ **Units:** Bandwidth is often expressed in Mbps/Gbps or Hz, whereas data rate is measured in bits per second (bps), Mbps, or MBps.
- ❑ Limited bandwidth increases distortion
- ❑ Have a direct relationship between data rate and bandwidth
 - Less bandwidth, more dispersion, and so less data rate

Bandwidth Vs Received Signal

Pulse before transmission

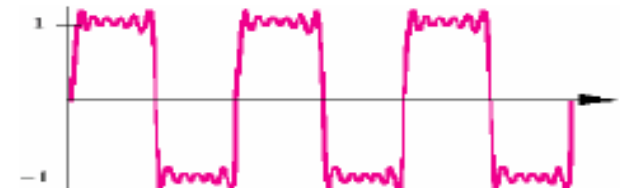
Bit rate: 2000 bits per second



Pulse after transmission



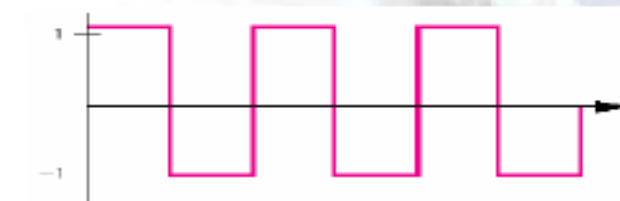
Bandwidth 500 Hz



Bandwidth 1700 Hz



Bandwidth 2500 Hz



Bandwidth 4000 Hz

Channel Capacity

- ❑ Channel capacity is the maximum rate at which information can be reliably transmitted over a communication channel, measured in bits per second (bps) or bits per symbol.
- ❑ It defines the absolute upper bound for error-free data transfer, determined by the [Shannon-Hartley theorem](#) based on bandwidth, signal power, and noise.

Nyquist Bandwidth

- ❑ Consider noise free channel
- ❑ Nyquist rate: $C = 2B$
 - Given bandwidth B , highest signal rate is $2B$
 - If rate of signal transmission is $2B$, then we can carry signal with frequencies no more than B
 - Given binary signal, data rate supported by B Hz is $2B$ bps
- ❑ Can be increased by using M signal levels
- ❑ $C = 2B \log_2 M$
- ❑ Limited by noise and other impairments

Analog Transmission

- ❑ Analog signal transmitted without regard to content
- ❑ May be analog or digital data
- ❑ Attenuated over distance
- ❑ Use amplifiers to boost signal
- ❑ Also amplifies noise

Digital Transmission

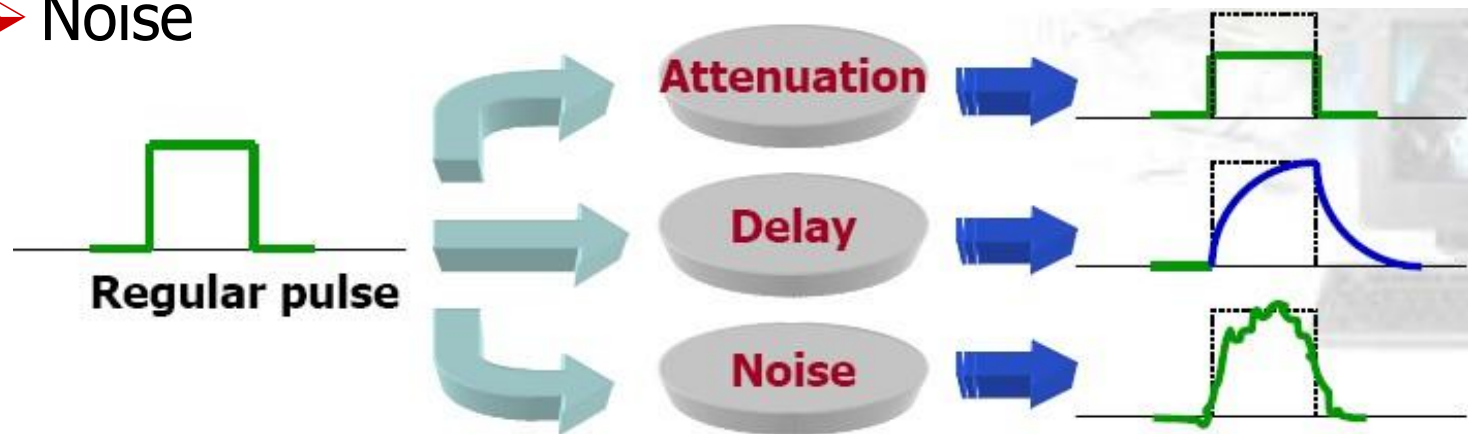
- ❑ Concerned with content
- ❑ Integrity endangered by noise, attenuation etc.
- ❑ Repeaters used
- ❑ Repeater receives signal
- ❑ Extracts bit pattern
- ❑ Retransmits
- ❑ Attenuation is overcome
- ❑ Noise is not amplified

Advantages of Digital Transmission

- ❑ Digital technology
 - Low cost LSI/VLSI technology
- ❑ Data integrity
 - Longer distances over lower quality lines
- ❑ Capacity utilization
 - High bandwidth links economical
 - High degree of multiplexing easier with digital techniques
- ❑ Security & Privacy
 - Encryption
- ❑ Integration
 - Can treat analog and digital data similarly

Transmission Impairments

- ❑ Signal received may differ from signal transmitted
 - Analog - degradation of signal quality
 - Digital - bit errors
- ❑ Caused by
 - Attenuation and attenuation distortion
 - Delay distortion
 - Noise



Attenuation

- ❑ Signal strength falls off with distance
- ❑ Depends on medium
- ❑ Received signal strength must be:
 - strong enough to be detected
 - sufficiently higher than noise to be received without error
- ❑ So increase strength using amplifiers/repeaters
- ❑ Attenuation is also an increasing function of frequency
 - So equalize attenuation across band of frequencies used
 - Eg. Using loading coils

Delay Distortion

- ❑ Only in guided media
- ❑ Propagation velocity varies with frequency
- ❑ Hence various frequency components arrive at different times
- ❑ Particularly critical for digital data
 - Since part of one bit spill over into others
 - Causing intersymbol interference

Noise (1)

- ❑ Additional signals inserted between transmitter and receiver
- ❑ Thermal
 - Due to thermal agitation of electrons
 - Uniformly distributed
 - White noise
- ❑ Intermodulation
 - Signals that are the sum and difference of original frequencies sharing a medium

Noise (2)

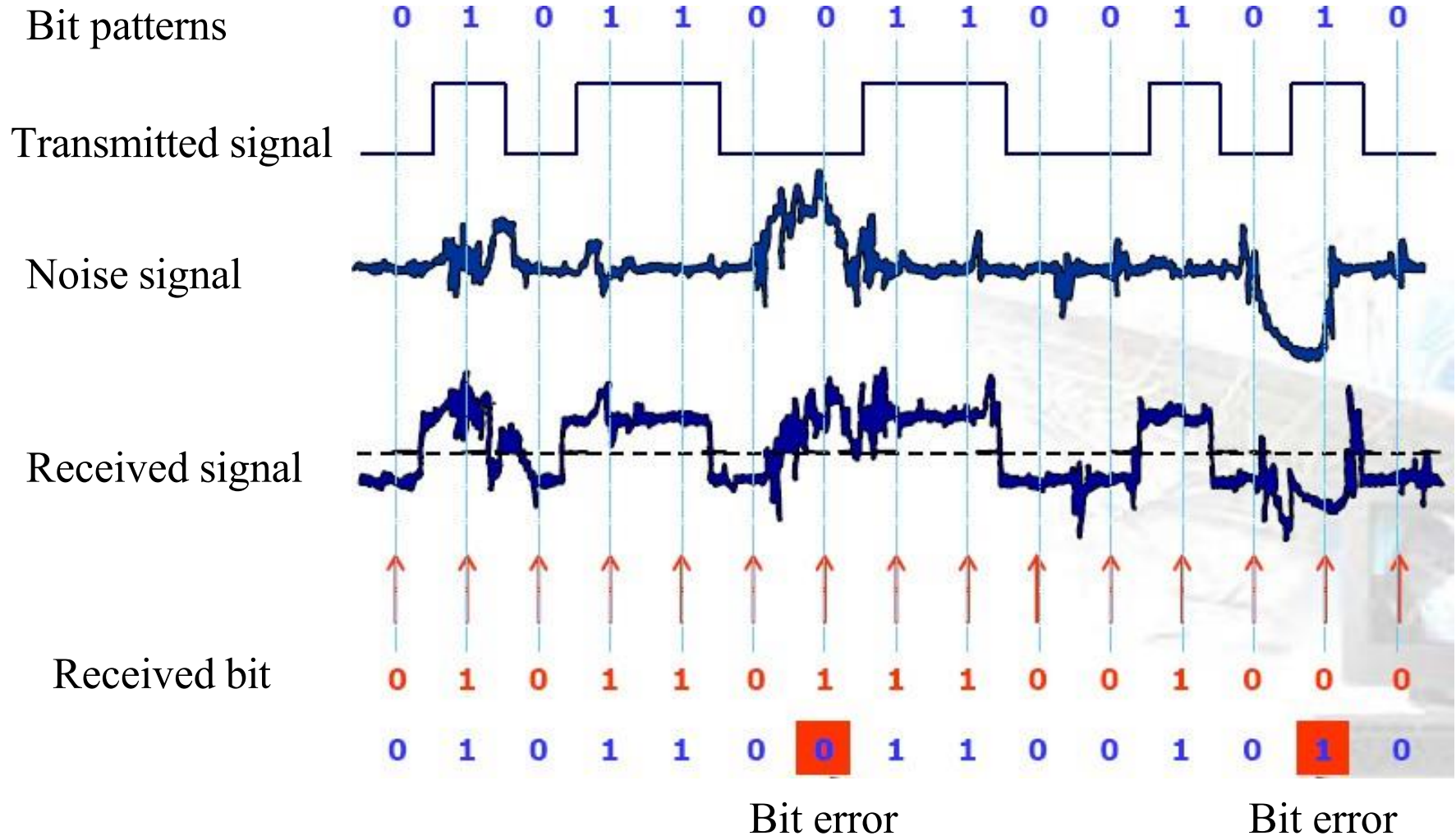
❑ Crosstalk

- A signal from one line is picked up by another

❑ Impulse

- Irregular pulses or spikes
 - e.g. External electromagnetic interference
- Short duration
- High amplitude
- A minor annoyance for analog signal
- But a major source of error in digital data
 - A noise spike could corrupt many bits

Noise (Example)



Channel Capacity

The maximum rate at which data can be transmitted over a given communication path, or channel, under given condition, is referred to as channel Capacity.

- ▶ **Data rate**
 - In bits per second
 - Rate at which data can be communicated
- ▶ **Bandwidth**
 - In cycles per second of Hertz
 - Constrained by transmitter and medium
- ▶ **Noise**
 - Average level of Noise over communication path
- ▶ **Error rate**
 - Rate at which errors occurs
 $1 \rightarrow 0$ or $0 \rightarrow 1$

Channel Capacity

- ❖ The greater the bandwidth of a facility, the greater the cost.
- ❖ All transmission channels are of limited bandwidth. Arise from physical properties or from deliberate limitations at the transmitter to prevent interference from other sources.

But–

- We want to get as high a data rate as possible at a particular limit of error rate for a given bandwidth.
- The main constraint on achieving this efficiency is noise.

Channel Capacity

- ▶ Consider noise free channel
- ▶ Nyquist rate: $C = 2B$
 - Given bandwidth B , highest signal rate is $2B$
 - If rate of signal transmission is $2B$, then we can carry signal with frequencies no more than B
 - Given binary signal, data rate supported by B Hz is $2B$ bps
- ▶ Can be increased by using M signal levels
- ▶ $C = 2B \log_2 M$
- ▶ Limited by noise and other impairments

Shannon Capacity Formula

- ❑ Consider data rate, noise and error rate
- ❑ Faster data rate shortens each bit so burst of noise affects more bits
 - At given noise level, high data rate means higher error rate
- ❑ Signal to noise ration (in decibels)
- ❑ $\text{SNR}_{\text{db}} = 10 \log_{10} (\text{signal/noise})$
- ❑ Capacity $C = B \log_2(1 + \text{SNR})$
- ❑ This is error free capacity

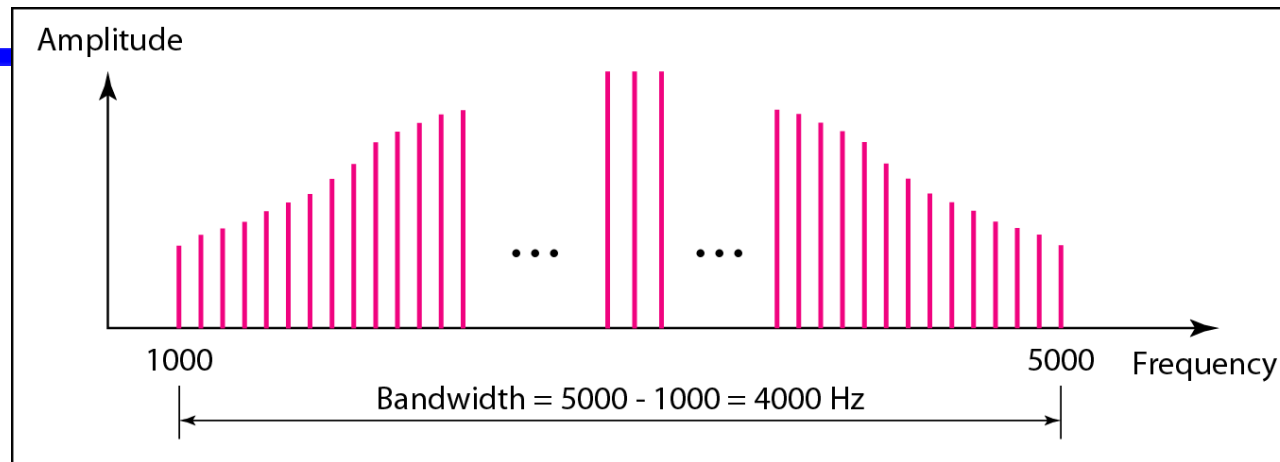
Discussion

- ✓ For a given level of noise, data rate increases if signal strength or bandwidth increases.
- ✓ But as signal strength increases, the effect of nonlinearities in the system, leading to an increase in intermodulation noise increases.
- ✓ The wider the bandwidth , the more noise is admitted because noise is assumed to be white.
$$N_o = kTB$$
- ✓ Thus as B increases, SNR decreases.

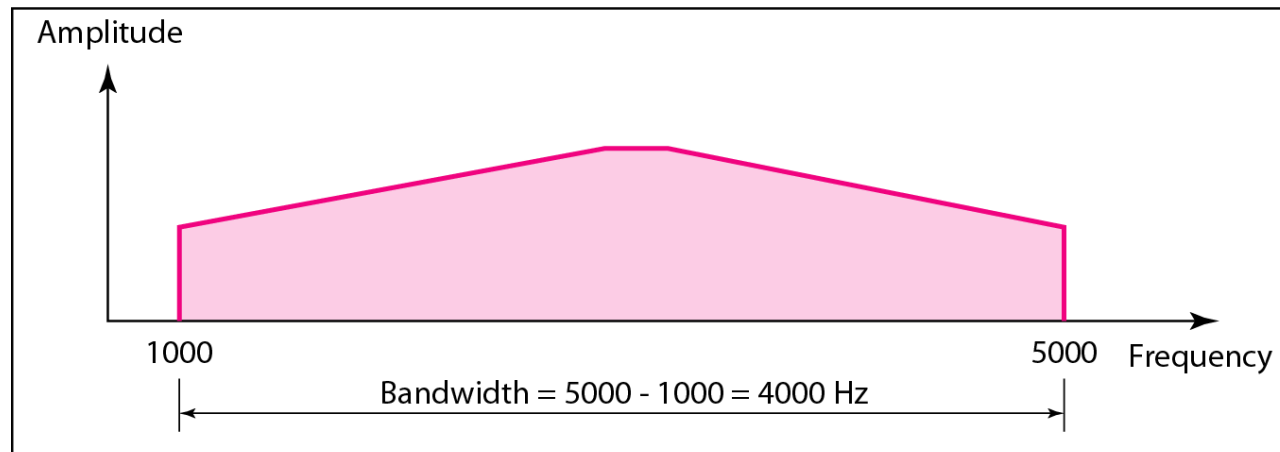
In communication systems, we commonly use periodic analog signals and nonperiodic digital signals.

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Figure *The bandwidth of periodic and nonperiodic composite signals*



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Example

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

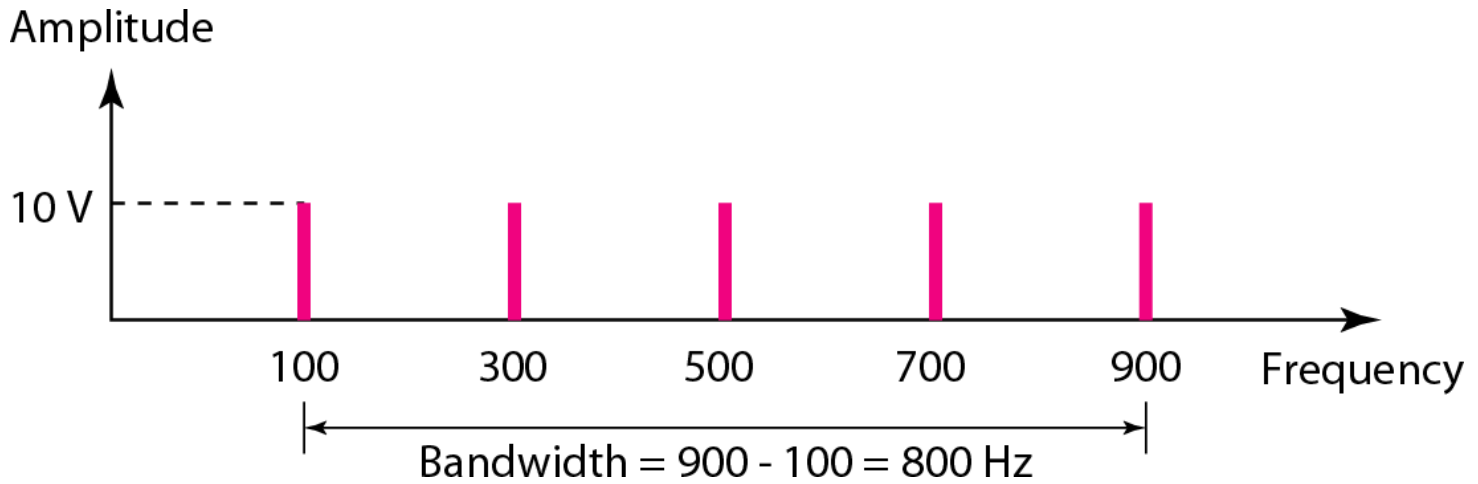
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz (see next Figure).

Figure *The bandwidth for Example*



Example

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

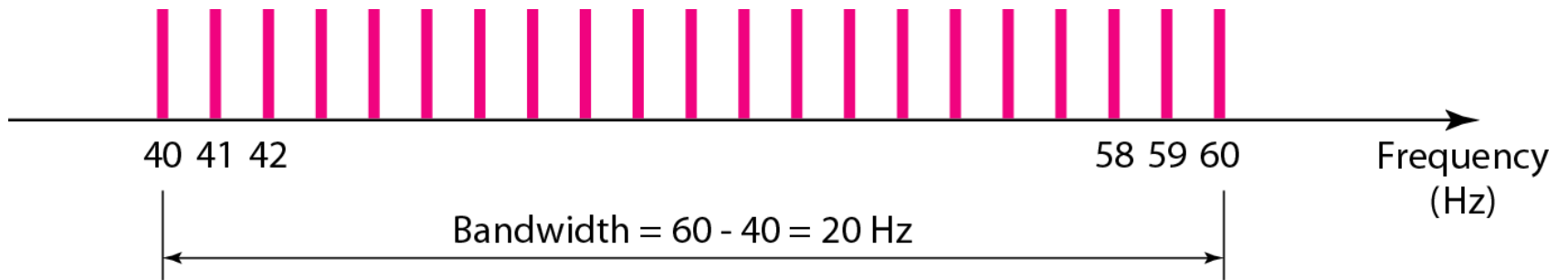
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \Rightarrow 20 = 60 - f_l \Rightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes (see next Figure).

Figure *The bandwidth for Example*





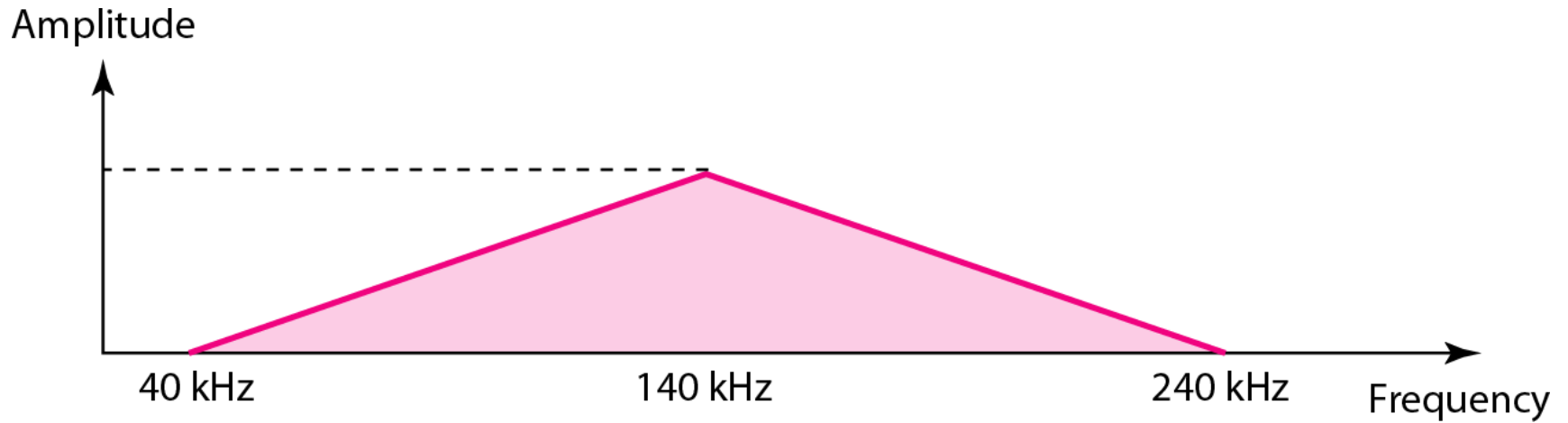
Example

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution

The lowest frequency must be at 40 kHz and the highest at 240 kHz. Next Figure shows the frequency domain and the bandwidth.

Figure *The bandwidth for Example*





Example

An example of a nonperiodic composite signal is the signal propagated by an AM radio station. Each AM radio station is assigned a 10-kHz bandwidth. The total bandwidth dedicated to AM radio ranges from 530 to 1700 kHz.



Example

Another example of a nonperiodic composite signal is the signal propagated by an FM radio station. Each FM radio station is assigned a 200-kHz bandwidth. The total bandwidth dedicated to FM radio ranges from 88 to 108 MHz.